THE DATA DRIVEN ZONE

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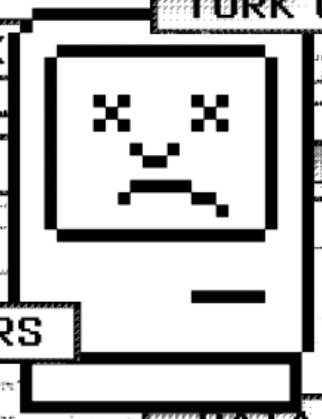
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THE DATA IS BEING ANNOTATED

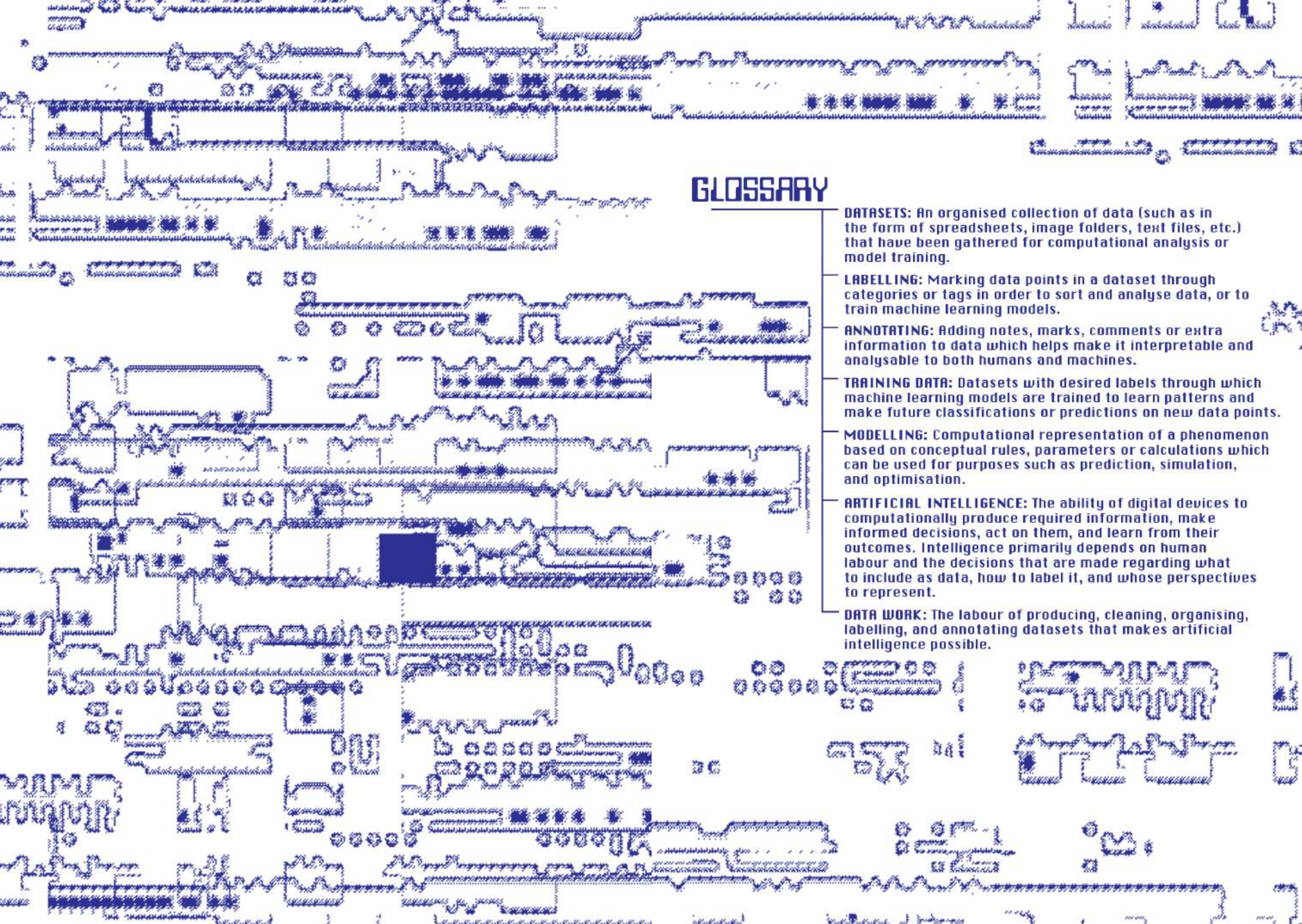
WHO KNOWS A MAN'S NAME, HOLDS THAT MAN'S LIFE IN HIS HANDS —
Ursula Le Guin, A Wizard of Earthsea

We prompted perplexity to give a list of academic metaphors used to refer to data work, for example, data coolie and plot it on a timeline with references. Next, we prompted it to identify metaphors used on the internet. This generated a list without references, and metaphors leaked between different kinds of roles working with data.



PROMPT MONKEYS
SPREADSHEET COWBOYS
AI UNDERCLASS / DIGITAL UNDERCLASS
QUANT / QUANTS
PENNY SQUIRRELS
NUMBER CRUNCHERS
AMAZON MTURK
DATA CONTROLLERS
DATA COOLIE
DATA CLEANERS
ETL ZOMBIES
GHOST WORKERS
DATA SERFS
DATA JANITORS
DATA WORKERS
MODEL BABYSITTERS
DATA GOVERNORS / DATA GOVERNANCE OFFICERS
ROCK-STAR DATA SCIENTISTS
CLICKWORKERS
QUERY SHAMANS
PIPELINE PLUMBERS
DATA POLICE
TURK GRINDERS
DATA PRODUCT OWNERS
DATA GRUNT
KNOWLEDGE WORK
DATA GATEKEEPERS
DATA MULE
DASHBOARD SERFS
TURKERS
API WRANGLERS
LABELING SQUIRRELS
MECHANICAL TURK
CONTENT MODERATORS
CROWD LABOUR
DATA ARTISTS
CROWDWORKERS
DATA HANDLERS
HUMAN COMPUTATION WORKERS
HIT GRINDERS
DATA ENTRY CLERKS
PIVOT TABLE WIZARDS
EXCEL MONKEYS
ANNOTATION FACTORY WORKERS
DATA FARMHANDS
SAFETY RATERS / AI SAFETY RATERS
SQL PEASANTS
DATA PLUMBERS
ALGORITHMIC PROLETARI[illegible]RM CAPITALISM)
CHART MONKEYS
DATA WRANGLERS
LABELERS / ANNOTATORS
DATA STEWARDS
DATA SHEPHERDS
DATA DUMPSTER DIVERS
DATA MAIDS / SPREADSHEET MAIDS
HUMAN RATERS / HUMAN LABELERS
DATA VAMPIRES
SEARCH QUALITY RATERS
DATA NINJAS
DATA WIZARDS / DATA MAGICIANS
DATA GURUS

It added that these metaphors "thrive in burnout memes and contrast academic gravity—purely pragmatic humour for coping with tool-specific hell".



GLOSSARY

DATASETS: An organised collection of data (such as in the form of spreadsheets, image folders, text files, etc.) that have been gathered for computational analysis or model training.

LABELLING: Marking data points in a dataset through categories or tags in order to sort and analyse data, or to train machine learning models.

ANNOTATING: Adding notes, marks, comments or extra information to data which helps make it interpretable and analysable to both humans and machines.

TRAINING DATA: Datasets with desired labels through which machine learning models are trained to learn patterns and make future classifications or predictions on new data points.

MODELLING: Computational representation of a phenomenon based on conceptual rules, parameters or calculations which can be used for purposes such as prediction, simulation, and optimisation.

ARTIFICIAL INTELLIGENCE: The ability of digital devices to computationally produce required information, make informed decisions, act on them, and learn from their outcomes. Intelligence primarily depends on human labour and the decisions that are made regarding what to include as data, how to label it, and whose perspectives to represent.

DATA WORK: The labour of producing, cleaning, organising, labelling, and annotating datasets that makes artificial intelligence possible.

Introduction

Sometime in the mid-2020s, a day of life in the South Indian city of Hyderabad — where there is also a place called Cyberabad, which combines all the technology townships that form the financial business districts called Hyderabad Information Technology and Engineering Consultancy City (HITEC City).

They are all located at the heart of a key geological feature, the Deccan Plateau. A dry, rocky plateau that is one of Earth's oldest and most stable land formations, consisting of granite and sedimentary rocks that are remnants of the supercontinent Gondwana.

Also, a day in Pune, a city in Western India, north of Hyderabad. Still in the Deccan.

For another, a day in Mumbai on the western coast, directly adjacent to the Deccan.

Or, a day in Ahmedabad, on the banks of the Sabarmati River.

Also, a day in Bergen, a 7,100 km-long straight line away from Deccan.

Some of us work full-time, sometimes 20 hours a day, for a meme stock company. Cleaning, labelling and annotating datasets for LLMs and autonomous vehicles.

Some of us volunteered precariously for a civic tech company struggling to keep datasets alive and meaningful.

Others work full-time for a university and a research institute, producing knowledge about what is to be done with AI.

What connects all of us?

We don't know how much we know about how these things work.

Where the data comes from. Where it goes when it is cleaned up, refined, written about.

All we have is something like folklore. Stories about something that happened with the latest software tool. A myth about another company. Proverbs about workload. Jokes about that one manager. Idioms to make our work quicker as we jump between projects.

This kind of folklore is often seen as imprecise, mystifying, or even counterproductive

What can folklore do to how we think with AI and algorithms?

This has been one of the key questions that haunts this zine.

Folklore can both produce and harbour a range of tangential affects.

These are often side-stepped by different master narratives of efficiency, responsibility, or care, struggling to counter each other, vying for our attention and participation.

In the following pages, you will find fragments from data factories in India where data workers with degrees in Computer Science, English, Human Resources and Business Analytics label and annotate large datasets to make them machine-readable. They are there because all other jobs in their domain are drying up. And it also pays well compared to everything else available on the job market.

Different users of AI have different stories to tell. From the viewpoint of data factories in India, the world looks something like this. Or at least it did from the limited perspective of the snapshot in time we present in this publication.

In the following pages, we combine the folk knowledge contributed by data factory workers with a selection of questions posed by a non-profit hate speech detection model, called Uli, in order to foreground the sophisticated processes of interpretation and sense-making involved in data work. These materials are integrated with noisy abstractions of datasets, interfaces, visualisations, and social media posts.

Putting together these materials helped us raise many questions; not all of them have been answered, and perhaps this is one of the purposes of this zine: to let them ring for someone else to answer.

Axe in the Forest

Several of us are gathered in the forest. But which one of us is chopping it all down?

It happened only recently. My father asked me to make something, some sort of video with AI. He asked me because, since I work with AI, I must know how to create a video. I'm like, I've seen people doing it, but I'm not the guy who cuts the tree.

I'm not the guy who cuts the tree. I'm the guy who makes the axe.

I trained to become a developer, and now I am working to become a café owner. The average developer is being chopped out of existence by the very axe I provide.

We often joke around, saying "OK, we're training this model to replace us..."

It has happened before.

The forest was shrinking, yet the Trees kept voting for the AXE AXE AXE AXE,

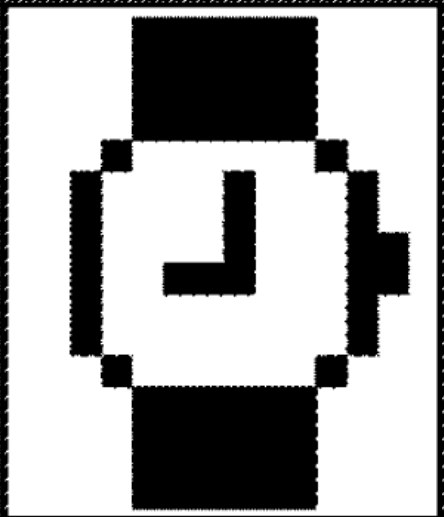
for the Axe was clever and convinced the trees that because his handle was made of wood, he was one of them.

QUESTION 1:

Can active participation create spaces where the axe can be wielded responsibly?

Murder

by AI



Do you know about the trolley problem?

It is absurd. You have the lever to divert the machine. You decide how many die. It is a thought experiment.

It is a man-made ethical conundrum.

Who should be hit?

There's a story that I've been too lazy to write. Sometimes we discuss it and laugh about it on Slack. The story of a Project Lead who wanted to murder a very particular enemy of his. He decides to meticulously influence the training of autonomous vehicles, over months and months of slow but steady labels and subtle strings and slight tweaks in the guidelines. Somehow (still need to figure out how to make it sound more realistic), he manages to make the system follow a certain set of guidelines (where the model performs as expected, but there's a trigger that sets in at the sight of the rarest of instances where this enemy comes across), which eliminates his arch-enemy.

QUESTION 2:

The promise of responsible ML/AI rests on an oft-quoted aphorism, "garbage in, garbage out". Models relying on low-paid, invisible labour will only work to shrink vibrant forests. Can a carefully organised chain of data labelling with relevant people and well-thought-out labels, and shared goals, produce something more relevant in which no one has to die or disappear?


Original audio
Follow
AI ka matalab
Reality , its coming
Ai
Bro forgot software engineers only create AI and what a software developer can do AI can't go on that level of customisation, It can only help to ease your work bt not
AI ?
it's
fun
a
giant
decline
but bro choose the truth
I mean, sure, it's fun at the beginning, but thank God something else is going to get to deal with all the deadlines set by shitty non-technical managers that don't know the difference between HTML and CSS.
problem now
215k
Dislike
Inner Voice of Every Corporate Emp
Everybody in the room is trying to quit.
#oyefoodiesingh #chandig
#chandigarh #foodblogge
#chandigarhblogger #viral
51w
AI vadapav nhi bna skta
Like Reply
coding
Lakhpati Vada Pav W
"And if you're one of those people that thinks all programmers will be taken over by AI, let me ask you a question. Did anyone actually enjoy coding in the first place?
And like, did anyone actually enjoy the LeetCode grinding for interviews that never came because the whole industry is in a giant decline? It's someone else's problem now. Some AI is going to have to deal with being your family's tech support from now on. And honestly, I'm happy for it. And me, I just get to go do something else, like maybe I'll become a plumber or something."
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The ghosts in the Machine

Data labelling: "Poorly labelled data leads to biased, unreliable outcomes"

How much training does one need to make sure that the data is labelled carefully?

There's no training; we just start working on the project from day one. We learn through reinforcement learning. Label, make mistakes, get feedback.

When labelling data for autonomous vehicles, we can say that most of the time, we are winging it.

"There have always been ghosts in the machine. Random segments of code have been grouped to form unexpected protocols. Unanticipated, these free radicals engender questions of free will, creativity, and even the nature of what we might call the soul"

Some of us are like ghost workers.
Some of us are the GHOSTS IN THE MACHINE.



QUESTION 3:

Can an annotator's role be reimagined? Where data labelling is not a devalued, replaceable role, but a role that requires careful work of making the deep messiness of life machine-readable"."

The speed of light

You and I were arguing about which was most beautiful.
The answer lay in the agreement rate.

What's the agreement rate?

A neural network cannot tell the difference between
a red ball and a blue ball

unless you and I label it for the network.
If we both label the same ball blue,
we get 100% agreement rate.

It can't be 100% agreement on most things.

My blue is different from yours.

In the data factory, we have a speed-of-light
way to reach the agreement rate. Usually, in
companies, what you do is run a lot of simulations.

Everyone does it, and you take the
average as an achievable speed and quality,
but in our company, it's called speed of light.

You take the best of the best,

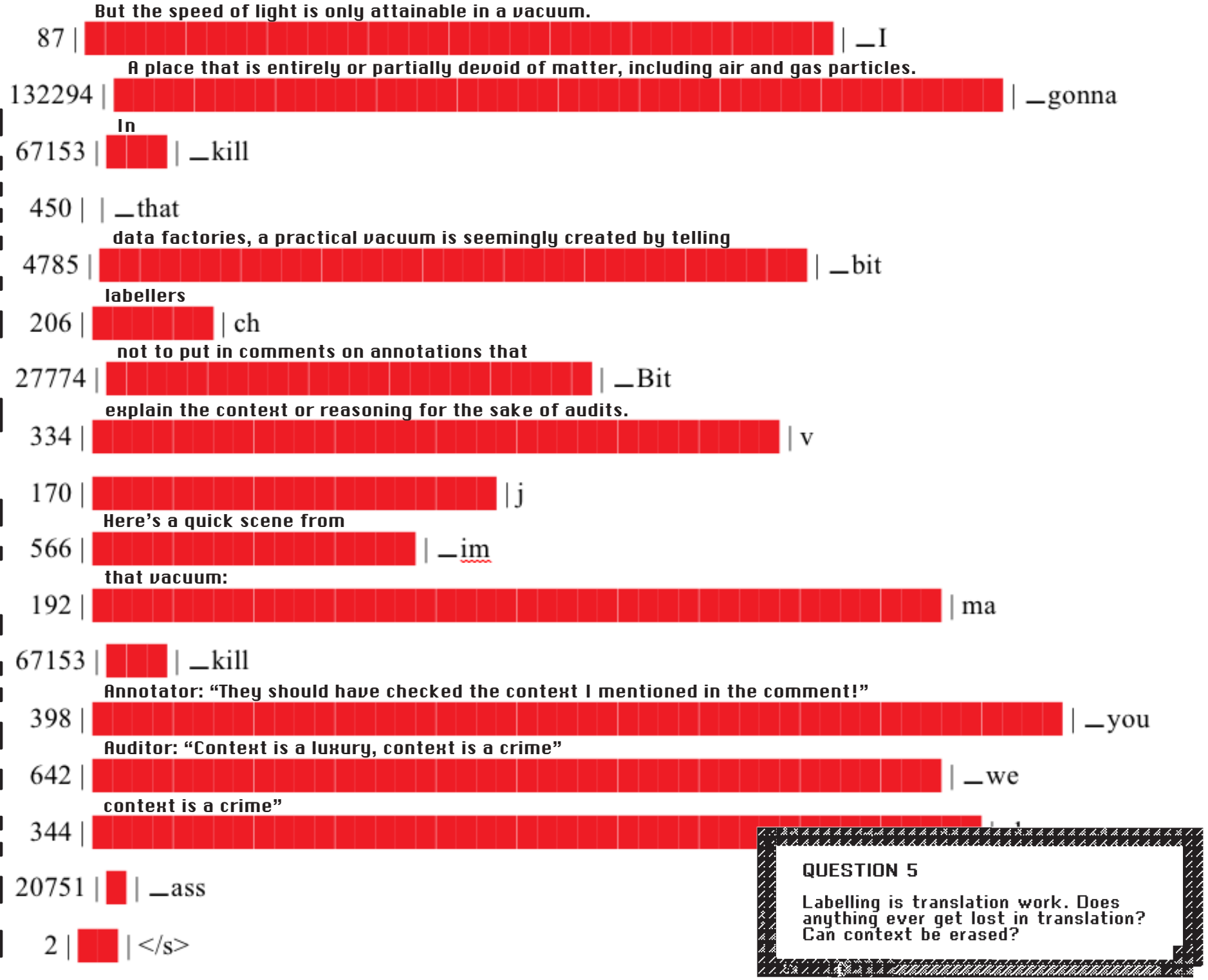
and that is what the target is going to be.

They brag about it. We do it the speed-of-light way.

QUESTION 4:

Can an active exchange of ideas, conversations, understanding of difference and disagreement produce more meaningful agreement rates?

Context is a luxury, context is a crime



Thank god it's Friday!

Not anymore, now it's the frustrated Friday.

What's a
frustrated Friday?

The day when performance emails are sent.
It tells me about the quality of my work.

And what happens if your
quality is anything less than
the established speed of light?

I get one week to
improve my
performance.

Frustrated Fridays

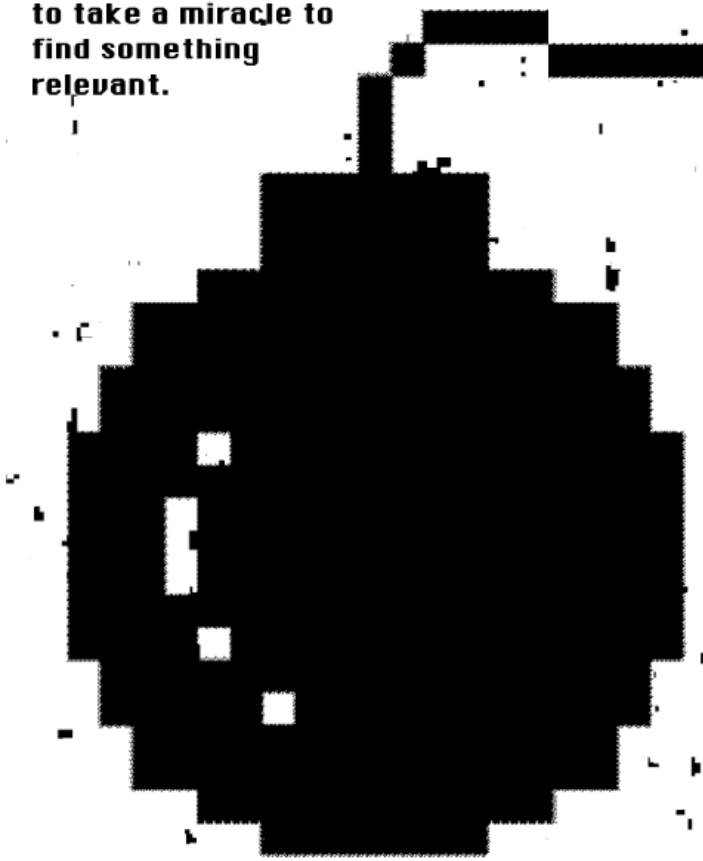
QUESTION 6

What will we get if we shift the question of how to label well to how people survive the mandate to label at all?

I'm trying so hard to quit

Everybody in the room is trying to quit.

Right now, the main reason not to move anywhere in the field is because of the job market. It is very unstable, and if you leave this job and try to find other jobs, it's going to take a miracle to find something relevant.



I know we are all saying we want to QUIT,

we want to QUIT,

but there

are no JOBS.

QUESTION 7

How do we organise so that participation in AI creates new spaces of trust rather than hierarchical layers of "bullshit" contributions?

David Graeber, *Bullshit Jobs: A Theory (2018)*

What does a data worker do?:

Autonomous vehicles labelling, analysing videos, with LiDAR / multi-camera views from different angles.

Dense captioning (video): labelling of videos and writing structured captions; Data workers are not allowed to use AI to write captions.

Image captioning: writing general descriptions of images; those in quality assurance (QA) check description quality. It requires a good command of English; at the industry level, only English datasets are being annotated in India.

Document/textbook structuring: standardising and capturing data in the form of tables for textbooks and diagrams, to convert them into a machine-readable form.

Creating questions and answers from multiple books, QA verifies the Q&A quality. An English graduate has to quickly learn to wrap their head around thermodynamics.

Cropping PDF files based on the question, finding the answer in the PDF, and drawing boxes so answer images can be collated (QA checks relevance).

Analysing an LLM's trace/"thinking" (chain-of-thought) by solving coding problems and comparing the model's reasoning to an "optimal" solution.

Comparing two responses and choosing which is better for users. Often referred to as Reinforcement Learning from Human Feedback (RLHF).

RLHF: comparing two responses and choosing which is better for users (strengths/weaknesses, verdict).

Editing/improving model responses based on noted omissions/issues. This is an extension of RLHF.

Robotics / AGI / egocentric data: labelling/working with videos of robots performing actions.
In the near future, data work could entail "physical AI" / robotics and more embodied data work performed to improve robots and real-world actions—captured through wearables and physically performing actions (running, lifting, sports motions) to capture training data.

